

Tufail Akram

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PROFESSIONAL SUMMARY

Accomplished Software Engineer with 5 years of expertise in backend and full-stack development, specializing in Python, FastAPI, Django, and scalable cloud architectures. Proven track record in engineering AI-powered solutions, including LLM integration, RAG pipelines, and autonomous AI agents to streamline complex workflows.

Leveraged deep expertise in **Snowflake**, **Snowpark**, and building **high-performance ETL pipelines** to optimize large-scale data transformation and processing. Skilled in AWS, Docker, Kubernetes, and CI/CD, with a focus on designing resilient microservices and production-grade systems. Collaborative developer with additional proficiency in React.js and Node.js for comprehensive full-stack delivery.

TECHNICAL SKILLS

Languages: Python, JavaScript, C, C++, Java

Frameworks: FastAPI, Django, React.js, Node.js

AI/ML: LLM, RAG, Machine Learning, AI Agents, AI Automation, AI Workflows

Data Engineering: Snowflake, Snowpark, ETL Pipelines, SQL

Cloud & DevOps: AWS, Docker, Kubernetes, Jenkins, CI/CD

Databases: PostgreSQL

Tools: Git, SonarQube, VPS

PROFESSIONAL EXPERIENCE

Software Engineer – BayRock Labs

Oct 2025 – Present | Remote

- Architected and scaled high-concurrency backend systems and REST APIs using **Python** and **FastAPI**, significantly improving system reliability and request throughput.
- Developed and deployed a specialized **Machine Learning (ML) model** in **Python** to autonomously determine and map "master" and "child" asset locations, leveraging rule-based algorithms to optimize location-intelligence accuracy.

- Engineered and deployed sophisticated **AI-driven** asset management platforms, utilizing advanced **LLM** and **RAG** pipelines for intelligent, context-aware data retrieval from large document sets.
 - Boosted operational efficiency by designing and deploying complex, autonomous **AI workflows** and multi-step automation agents that streamlined critical business processes.
 - Built robust, high-performance **ETL pipelines** using **Snowflake** and **Snowpark** with complex **SQL** to optimize large-scale data transformation and processing.
 - Orchestrated containerized deployments across Docker and Kubernetes, integrating automated CI/CD pipelines for secure, rapid, and error-free application delivery.
 - Integrated essential AWS cloud services and location-intelligence geolocation APIs to enhance platform capabilities and provide accurate tracking services.
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Full Stack Python Developer – Fin Infocom Pvt Ltd

Oct 2024 – Oct 2025 | Hyderabad

- Spearheaded the development of an enterprise-grade POS system, automating billing, inventory management, and financial reporting.
 - Secured cloud infrastructure by implementing serverless AWS Lambda APIs governed by strict IAM roles and security policies.
 - Optimized backend performance, achieving a significant reduction in API latency and ensuring stability under peak user loads.
 - Developed a modular microservices architecture to support long-term scalability and maintainability of enterprise applications.
 - Optimized data retrieval and system reporting by writing high-performance **SQL** queries against the PostgreSQL database.
 - Integrated a data pipeline to replicate POS data into a **Snowflake** warehouse for advanced business intelligence reporting.
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Full Stack Developer – KozyKreative Concepts Pvt Ltd

May 2023 – Oct 2024 | Hyderabad

- Engineered "OptionPerks," a high-frequency algorithmic trading platform featuring low-latency real-time data processing.
- Launched a comprehensive medical consultancy portal with integrated real-time video consultation and appointment features.
- Delivered end-to-end full-stack solutions utilizing React.js, Django, and PostgreSQL for diverse client requirements.
- Facilitated seamless communication by integrating third-party WebRTC and messaging APIs.

Backend Developer (Contract Role)

Aug 2021 – Apr 2023

- Developed robust backend services and RESTful APIs to support various data-intensive client applications.
 - Enhanced data integrity and retrieval speed through optimized PostgreSQL schema design and indexing.
 - Implemented automated ETL pipelines to facilitate seamless data migration and processing across multiple environments.
 - Modernized legacy codebases, resulting in improved system throughput and a more resilient backend architecture.
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PROJECTS

ML-Driven Geolocation Intelligence Platform

Engineered a specialized **Machine Learning (ML) model** in **Python** to autonomously determine and map "master" and "child" asset locations, leveraging complex rule-based algorithms for optimization.

Built location-based services using AWS geolocation APIs and implemented core tracking/mapping features.

AI Workflow-Driven Customer Support System

Built an AI-powered support system in **Python**, utilizing advanced **LLM** and **RAG** architecture for intelligent, context-aware data retrieval.

Implemented intelligent workflows and multi-step AI agents for automated ticket handling.

Developed high-performance APIs using FastAPI.

AI Agent Automation System

Developed advanced, autonomous **AI Agents** in **Python** for complex workflow automation and decision-making.

Integrated **LLM** for intelligent processing.

OptionPerks – Algorithmic Trading Platform

Developed an algorithmic trading platform featuring low-latency real-time data processing using **Python**.

Implemented the core execution engine, laying the foundation for integrating future **Machine Learning** models for predictive analytics.

Integrated financial APIs and built the execution engine.

Data Pipeline and ETL System

Built robust, high-performance **ETL pipelines** using **Snowflake** and **Snowpark**.

Automated large-scale data ingestion and transformation workflows using complex **SQL** for optimization.

Medical Consultancy Platform

Built a telemedicine platform with integrated real-time video consultation and appointment features using React.js, Django, and PostgreSQL.

Designed a secure backend for patient data management.

POS Enterprise System

Developed comprehensive billing, inventory, and financial reporting modules using Python and Django.

Optimized data retrieval and enhanced system performance and scalability.

EDUCATION

Bachelor of Computer Applications (BCA)
NIIS Institute of Business Administration
2018 – 2021

ACHIEVEMENTS

- **Technical Leadership:** Directed the development of multiple high-impact cloud-native and AI-integrated platforms.
- **Innovation:** Pioneered autonomous AI agents that reduced manual task handling by automating complex business workflows.
- **Performance Optimization:** Successfully increased API response times and system scalability through architectural refinements.
- **DevOps Excellence:** Reduced deployment errors and manual overhead by implementing end-to-end CI/CD automation.